Hierarchical Label Smoothing and a P2 Detection Head for Vietnamese Traffic Sign Detection: A Controlled Negative-Result Study

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***Abstract*—Vietnamese traffic sign detection using the QCVN 41 regulatory taxonomy has not previously been studied with a loss-level hierarchical supervision signal and an architecture-level small-object detection head evaluated independently and under a leakage-free protocol. This paper investigates two such interventions against a flat-label YOLO26n baseline on the VNTS dataset (3,216 images, 8,334 instances, 52 classes): (1) Hierarchical Label Smoothing (HLS), which redistributes classification-target probability mass across QCVN 41 category siblings, with hyperparameters selected via 3-fold cross-validation restricted to the training pool; and (2) a P2 detection head added at stride 4 for improved small-object recall. Under a corrected evaluation protocol in which the test set is used exactly once after all hyperparameter and checkpoint decisions are finalized, neither intervention improves on the flat baseline. HLS's best cross-validated configuration is statistically indistinguishable from the baseline at every frequency tier (Wilcoxon signed-rank, all $p > 0.05$) and trails it on both global metrics and the Rare tier, even though the classification-loss weight λ_{cls} produces a large, low-variance effect within the HLS family itself. The P2 head produces a clear, statistically significant negative result ($p = 3.4 \times 10^{-7}$ globally), with a precision/recall/mAP signature consistent with poorly calibrated confidence in the newly initialized branch rather than a general loss of detection ability. We report both results in full, including the earlier interim result they supersede, and discuss what the corrected protocol implies for hierarchical loss design on small, imbalanced, regulator-defined taxonomies.**

***Index Terms*—traffic sign detection, hierarchical label smoothing, YOLO, small-object detection, class imbalance, Vietnamese traffic signs, QCVN 41**

I. Introduction

A. Problem Definition

Traffic sign detection is the task of localizing and classifying traffic signs within road-scene images. This work addresses Vietnamese traffic sign detection specifically, using the Vietnamese Traffic Signs (VNTS) dataset, which contains 3,216 images and 8,334 annotated instances across 52 sign classes governed by the QCVN 41 national road signage regulation.

B. Motivation

Existing Vietnamese traffic sign datasets and models treat all sign classes as independent ("flat-label"), ignoring the hierarchical regulatory structure defined by QCVN 41, under which every sign belongs to one of five broad categories: Prohibitory, Warning, Mandatory, Information, and Supplementary. This structure is a natural and previously unused source of supervision. The dataset also exhibits severe class imbalance, with instance counts ranging from 3 to 1,071 per class, which is representative of real deployment conditions but makes rare-class detection difficult under standard training.

This paper investigates whether exploiting the QCVN 41 hierarchy through the training loss improves detection accuracy, particularly for underrepresented sign classes, and separately whether an architecture-level small-object detection enhancement (a P2 detection head) provides an independent benefit. To our knowledge, these two questions have not previously been studied together, nor evaluated under a protocol that fully separates hyperparameter selection from test-set evaluation, for Vietnamese traffic signs.

C. Applications

- Advanced Driver Assistance Systems (ADAS) — real-time sign recognition for driver alerts.
- Autonomous vehicle perception stacks operating in Vietnamese road environments.
- Automated road-infrastructure auditing, e.g., verifying signage compliance with QCVN 41.

D. Scope

The system takes a single RGB road-scene image (resized to 640×640) and outputs a set of bounding boxes, each with a predicted class among the 52 QCVN 41 sign identifiers. The scope is detection only, on single frames; tracking, depth estimation, real-time embedded deployment, OCR of sign text, and multi-camera fusion are out of scope.

E. Approach Overview

Three models are trained and compared, isolating two independent design questions against the same flat-label baseline:

- **Model B (baseline):** unmodified YOLO26n, standard training procedure.
- **Model A (loss-level intervention):** architecturally identical to Model B, trained with Hierarchical Label Smoothing (HLS), which injects the QCVN 41 taxonomy into the classification-loss target distribution without modifying

the model architecture. HLS's hyperparameters (target-gap width, classification-loss weight) are selected via cross-validation on the training pool, not fixed a priori and not selected against the test set.
- **Model P2 (architecture-level intervention):** YOLO26n augmented with an additional P2 detection head for higher-resolution small-object detection, trained with the same flat loss target as Model B, isolating architecture as the sole variable in that comparison.

Unlike prior hierarchical detection work, which typically requires additional detection-head branches or multi-stage cascades to carry the hierarchical signal, the hierarchical (HLS) and architectural (P2) interventions here are tested independently of one another, each against the same control.

II. Related Works

Lu *et al.* [1] proposed a two-stage hierarchical traffic sign recognition framework that first predicts broad sign shape categories before identifying specific sign classes, demonstrating that coarse-to-fine prediction improves accuracy over flat-label approaches. This work similarly exploits a coarse-to-fine structure, but implements it through the loss function of a single-stage detector rather than a sequential two-stage pipeline, preserving inference speed.

Zhang *et al.* [2] incorporated hierarchical semantic relationships into a traffic sign detection framework via structured category-level feature learning. This work tests the same underlying principle at the loss level rather than the architecture level.

Usmani *et al.* [4] proposed a hierarchical YOLOv8-based detector with real-time text recognition for UAE traffic signs. Their hierarchy is defined by sign shape and text content for a UAE-specific sign system, and their study does not investigate the sensitivity of hierarchical supervision to loss-weighting hyperparameters — a factor this work identifies as critical.

Tsenkova *et al.* [5] introduced hYOLO, a hierarchical extension of YOLOv8 with a modified hierarchical loss function. Their reported finding that increasing classification-loss weight improves hierarchical YOLO performance motivated the classification-weight ablation conducted in this work; the same qualitative pattern (higher weight helps within the HLS family) is observed here, though the improvement does not carry through to a significant gain over a flat baseline.

Zhu *et al.* [7] introduced TPH-YOLOv5, which adds a dedicated high-resolution detection head ("P2", stride 4) sourced from an early backbone stage, extending detectable object size down to roughly 4×4 pixels, specifically to improve small- and tiny-object recall on drone-captured imagery. Since VNTS sign instances are uniformly small relative to image resolution, this work tests whether the same architectural enhancement transfers to traffic sign detection, independent of the loss-level HLS intervention.

Ertler *et al.* [3] introduced the Mapillary Traffic Sign Dataset, one of the largest traffic sign benchmarks available at global scale. It does not include a taxonomy aligned with Vietnamese QCVN 41 regulations, motivating the use of VNTS despite its smaller scale.

Sapkota *et al.* [6] detail the architectural enhancements of YOLO26, including NMS-free end-to-end inference and the MuSGD optimizer, which form the base architecture used in this work.

Genermont *et al.* [8] investigate hierarchical novelty detection for traffic sign recognition, motivating the Hierarchical Cosine Loss direction identified as future work in this paper.

III. Data

A. Dataset Overview

The VNTS dataset is a pre-existing, publicly available dataset obtained via Kaggle, with annotations provided in YOLO format. No manual re-annotation was performed; the QCVN 41 category mapping was derived programmatically from a deterministic lookup table built from the QCVN 41 regulation text. Table I summarizes the dataset.

TABLE I
DATASET OVERVIEW

Property	Value
Total images	3,216
Total annotated instances	8,334
Sign classes	52
Train / test split	2,552 / 639
Average objects per image	2.59

B. Class Imbalance and Frequency Tiers

Instance counts per class range from 3 to 1,071, forming a severe long-tail distribution. Classes were stratified into three tiers using the 25th and 75th percentiles of the instance-count distribution as cutoffs (Table II).

TABLE II
FREQUENCY TIERS

Tier	Threshold	Classes	Instances
Frequent	$\geq$100	24	7,201
Medium	30–99	15	873
Rare	$<$30	13	260

C. QCVN 41 Hierarchical Taxonomy

All 52 classes were mapped to five QCVN 41 Level-1 categories via a deterministic lookup table (Table III). Two label inconsistencies in the raw dataset were identified and resolved during this process: a class labeled "B.8a" was identified as P.108a (a Prohibitory sign) based on visual inspection, and a class labeled "Camera" (referring to speed-camera warning signs, not a standard QCVN 41 code) was assigned to the Information category as a practical decision.

TABLE III
QCVN 41 CATEGORY DISTRIBUTION

Category	Classes	Instances	% of Total
Prohibitory	23	5,147	61.8%
Warning	18	1,727	20.7%
Mandatory	4	798	9.6%
Information	4	472	5.7%
Supplementary	3	190	2.3%

D. Spatial Characteristics

Median bounding-box area remains under 0.25% of image resolution across all three frequency tiers (Frequent 0.14%, Medium 0.24%, Rare 0.19%), confirming that object scale does not confound tier-based performance comparisons: signs are uniformly small regardless of class frequency. This uniform smallness motivates testing the P2 architectural enhancement: if a meaningful fraction of instances fall near or below the effective resolution of YOLO26n's standard highest-resolution head (P3, stride 8), an additional stride-4 head could plausibly help regardless of class frequency.

IV. METHODOLOGY

A. Base Architecture

All three models build on YOLO26n: a CSP-based convolutional backbone, PAN-FPN multi-scale neck, and an anchor-free, NMS-free end-to-end detection head. Models A and B use the architecture unmodified (2,384,976 parameters, 5.2 GFLOPs) and are architecturally identical to one another; the only difference between them is the classification-loss target. Model P2 adds a fourth, higher-resolution detection head, increasing parameter count to 2,429,832 (+1.9%) and compute to 6.8 GFLOPs (+30.8%).

B. Model B — Flat-Label Baseline

Model B uses YOLO26n's standard training procedure without modification. The classification target at a positive anchor is not a literal one-hot vector: YOLO26n's Task-Aligned Assigner (TAL) scales the target for the assigned ground-truth class by a box-quality metric (prediction confidence combined with IoU), so that poorly localized predictions receive a lower target even for the correct class. Critically, all other 51 classes remain exactly zero — TAL performs quality-scaled confidence adjustment for a single class, not cross-class redistribution.

C. Model A — Hierarchical Label Smoothing (HLS)

Let $S(c)$ denote the set of sign identifiers sharing class c's QCVN 41 Level-1 category, including c itself. The HLS target redistributes probability mass across three groups:

- the correct class receives $1 - \alpha - \beta$;
- each of the $|S(c)| - 1$ sibling classes within the same category receives $\beta/(|S(c)| - 1)$;
- each of the remaining classes outside the category receives $\alpha/(N - |S(c)|)$, where $N = 52$.

This smoothed target replaces the TAL-scaled target at positive anchors inside the standard binary cross-entropy classification loss; the bounding-box regression and distribution focal loss components are unchanged. HLS is applied identically to both the one-to-many and one-to-one assignment heads of YOLO26n's end-to-end loss. This substitution fixes the correct-class target at $1 - \alpha - \beta$ regardless of box localization quality, discarding the TAL quality-scaling signal that Model B retains.

As a concrete illustration, for a class c in the Warning category ($|S(c)| = 18$), under Model A's final hyperparameters ($\alpha = 0.05$, $\beta = 0.10$, $N = 52$): the correct class receives $1 - 0.05 - 0.10 = 0.85$; each of the other 17 Warning-category classes receives $0.10/17 \approx 0.0059$; each of the remaining 34 classes outside Warning receives $0.05/34 \approx 0.0015$.

The $(\alpha, \beta, \lambda_{cls})$ hyperparameters used for Model A's final training were not fixed in advance; they are the winning configuration selected via the cross-validation ablation described in Section IV-E and reported in Section V-B.

D. Model P2 — Architectural Small-Object Enhancement

Model P2 adds a dedicated high-resolution detection head at stride 4 ("P2"), following the extra-head design used in TPH-YOLOv5 for small- and tiny-object detection. P2 features are sourced from an earlier backbone stage than YOLO26n's default earliest head (P3, stride 8) and fused with upsampled P3 features in the neck, extending the range of reliably detectable object sizes downward. Model P2 is initialized from the pretrained `yolo26n.pt` checkpoint wherever layer shapes match; the new P2-specific layers are randomly initialized, since no pretrained weights exist for them. Model P2 uses the same flat (non-hierarchical) classification target as Model B, so architecture is the sole independent variable in that comparison.

E. Training Configuration

Training proceeds in two phases, structured to keep the 639-image held-out test set completely unused until every hyperparameter and checkpoint decision has already been made.

Phase 1 — HLS hyperparameter selection (Model A only). Four HLS configurations (varying α, β, and classification-loss weight λ_{cls}) are evaluated via 3-fold multilabel-stratified cross-validation, restricted entirely to the 2,552-image training pool. Each fold-configuration combination trains for 45 epochs, a budget set by Kaggle's per-session runtime limit (12 hours) and weekly GPU quota (30 hours); the configuration with the highest mean cross-validation mAP@0.5 is selected as Model A's final configuration.

Phase 2 — Final training (all three models). The winning HLS configuration (Model A), the flat baseline (Model B), and the P2-head baseline (Model P2) are each trained for 100 epochs on an identical split: approximately 90% of the training pool for training, and a stratified internal validation carve-out (~10%) used only for best-checkpoint selection during training. All three final models are trained and evaluated on exactly the same split, generated once and reused across all three.

Test evaluation. The 639-image test set is evaluated exactly once per final model, after training and checkpoint selection are complete. No configuration, hyperparameter, or checkpoint decision anywhere in the pipeline is made by looking at test-set performance.

Table IV summarizes the training configuration; Table V summarizes the three final models.

TABLE IV
TRAINING CONFIGURATION

Parameter	Value
Base weights	yolo26n.pt (COCO-pretrained)
Input resolution	640×640
CV ablation epochs (per fold)	45
CV folds	3 (multilabel-stratified)
Final training epochs	100
Internal validation carve-out	$\sim$10% of training pool
Batch size	32
Optimizer	AdamW (auto-selected)
Augmentation	Mosaic, HSV jitter, h-flip
Random seed	0 (fixed across all runs)
Hardware	NVIDIA Tesla T4 (Kaggle)

TABLE V
FINAL MODELS

Model	Architecture	Loss target
B	YOLO26n (unmodified)	Flat (TAL-scaled one-hot)
A	YOLO26n (unmodified)	HLS: α=0.05, β=0.10, λ_{cls}=1.5
P2	YOLO26n + P2 head	Flat (TAL-scaled one-hot)

Because HLS modifies only the classification target via a single lookup into a precomputed 52×52 smoothing matrix per batch, it adds negligible training overhead and has no effect on inference: at inference time, Model A and Model B execute an identical forward pass through an identical architecture. Inference cost is therefore identical between Models A and B by construction. Model P2's inference cost differs from both by construction, due to its additional detection head.

F. Note on an Earlier Interim Result

An earlier interim baseline figure (mAP@0.5 = 0.9487, mAP@0.5:0.95 = 0.7171) was obtained under an earlier evaluation protocol that used the test set as the validation split during training. That protocol was revised prior to the results reported here, so the interim figure is not directly comparable to the results below and is superseded by Section V-B.

G. Evaluation Metrics

- Precision, Recall (macro-averaged across classes)
- mAP@0.5 and mAP@0.5:0.95 (standard COCO-style detection metrics)
- Per-class AP50, stratified by frequency tier (Frequent / Medium / Rare)
- Wilcoxon signed-rank test on paired per-class AP50 scores (Model A vs. Model B, Model P2 vs. Model B), computed once on the held-out test set after all hyperparameter and checkpoint decisions are finalized

V. RESULTS

A. HLS Hyperparameter Selection (Cross-Validation)

Four HLS configurations were evaluated via 3-fold multilabel-stratified cross-validation, 45 epochs per fold, restricted to the training pool (Table VI). Within the HLS family, classification-loss weight (λ_{cls}) is the dominant driver of performance: both $\lambda_{cls} = 1.5$ configurations substantially outperform both $\lambda_{cls} = 0.5$ configurations at every tier, most dramatically in the Rare tier (0.37–0.40 vs. 0.19–0.29). Widening the target gap alone, without a correspondingly higher classification weight, degrades performance rather than helping. HLS-3_hicls ($\alpha = 0.05$, $\beta = 0.10$, $\lambda_{cls} = 1.5$) achieved the highest mean cross-validation mAP@0.5 at every tier and was selected as Model A's final configuration. This λ_{cls} effect is both large and low-variance — fold-level standard deviations stay within roughly 0.005–0.03 across all four configurations — indicating a real, reproducible effect within the HLS family.

TABLE VI
HLS CROSS-VALIDATION ABLATION (MEAN $\pm$ STD ACROSS FOLDS)

Config	α	β	λ_{cls}	Global	Freq	Rare
HLS-1	0.05	0.10	0.5	0.579$\pm$.005	0.752$\pm$.009	0.285$\pm$.013
HLS-2_wide	0.01	0.20	0.5	0.527$\pm$.009	0.732$\pm$.005	0.193$\pm$.020
HLS-3_hicls*	0.05	0.10	1.5	0.687$\pm$.008	0.808$\pm$.006	0.403$\pm$.023
HLS-4_combo	0.01	0.20	1.5	0.656$\pm$.005	0.796$\pm$.002	0.370$\pm$.017

Robustness within the family is a separate question from whether the winning configuration beats the flat baseline; Section V-B shows it does not.

B. Final Test-Set Results

Each model was evaluated once on the 639-image held-out test set, after training and checkpoint selection were complete (Table VII, Table VIII). Model A is close to Model B at every tier: it edges ahead of Model B on Frequent and Medium by less than one percentage point, but trails on both global metrics and, most notably, on the Rare tier (0.8053 vs. 0.8649) — the same tier where the CV ablation showed λ_{cls} having its largest effect. The honest reading is therefore not "HLS does nothing": within the HLS family, λ_{cls} matters a great deal and reproducibly, but even the best-tuned configuration, trained and tested once under the corrected protocol, does not clear the bar the flat baseline already sets. Model P2 underperforms Model B substantially at every tier and metric, with the largest gap in Recall (0.6708 vs. 0.8796).

C. Statistical Comparison vs. Baseline

A Wilcoxon signed-rank test on paired per-class AP50 scores was computed once on the test-set results above (Table IX). Model A shows no statistically significant difference

TABLE VII
FINAL TEST-SET RESULTS

Metric	Model B	Model A	Model P2
Precision	0.8337	0.8295	0.8455
Recall	0.8796	0.8689	0.6708
mAP@0.5	0.9257	0.9131	0.8335
mAP@0.5:0.95	0.6945	0.6690	0.6173

TABLE VIII
MAP@0.5 BY FREQUENCY TIER

Tier	Model B	Model A	Model P2
Frequent	0.9536	0.9556	0.8839
Medium	0.9336	0.9387	0.8523
Rare	0.8649	0.8053	0.7188

from Model B at any tier (all $p > 0.05$): there is no evidence that the CV-selected HLS configuration improves on the flat baseline. Model P2, by contrast, differs significantly from Model B at the Global, Frequent, and Medium tiers ($p < 0.05$); the Rare tier approaches significance ($p = 0.0645$) in the same negative direction. Model P2's difference from baseline is a clear, statistically supported negative result, not a null one.

TABLE IX
WILCOXON SIGNED-RANK TEST (p-VALUES)

Tier	n	A vs. B	P2 vs. B
Global	52	0.5539	3.4×10^{-7}
Frequent	24	0.8774	3.6×10^{-7}
Medium	15	0.8139	0.0131
Rare	13	0.2439	0.0645

D. Per-Category Analysis (Model A vs. Model B)

Per-class AP50 deltas (Model A minus Model B) were grouped by QCVN 41 category, to test whether category size predicts which classes benefit from HLS (Table X). The category-size hypothesis does not survive contact with the taxonomy's sample sizes. Supplementary has only 3 classes and 190 instances total, so a single confused or misclassified class can swing its category-level mean substantially; its large negative delta (-0.153) is at least as consistent with sampling noise from a 3-class category as with a genuine categorical effect, and the same caveat weakens confidence in the small positive deltas for Mandatory and Information (4 classes each). The two largest categories (Warning, 18 classes; Prohibitory, 23 classes) are the only ones large enough for their mean delta to be a reasonably stable estimate, and both show small losses rather than gains. A Pearson correlation between raw per-class training-instance count and AP50 delta was negligible and non-significant ($r = 0.0737$, $p = 0.6037$), consistent with class rarity not predicting HLS's effect either.

TABLE X
MEAN AP50 DELTA BY CATEGORY (MODEL A − MODEL B)

Category	Classes	Mean AP50 Delta
Supplementary	3	-0.1527
Mandatory	4	$+0.0281$
Information	4	$+0.0344$
Warning	18	-0.0093
Prohibitory	23	-0.0120

E. Architecture-Only Comparison: Model P2

Model P2's failure has a specific signature, not a diffuse one. Precision (0.8455) is the highest of all three models, above both Model A and Model B, while Recall collapses to 0.6708, and mAP@0.5 — a threshold-independent metric that integrates over the full precision-recall curve, not just one operating point — also drops clearly, from 0.9257 (Model B) to 0.8335. This combination rules out the simplest interpretation, that Model P2 got worse at detecting and classifying signs across the board: a model with a genuine localization or classification failure would not simultaneously post the highest precision of the three. Instead, the pattern points to poorly calibrated confidence scores from the new, untrained P2 branch — many of the model's detections are plausibly correct but sit just below whatever confidence threshold Recall is measured at, and the mAP@0.5 drop shows this is not an artifact of one threshold choice.

The most likely cause of that miscalibration is that Model P2 introduces new, randomly initialized parameters (the P2 head and its fusion path) that receive no benefit from COCO pretraining and must learn confidence calibration from scratch within a 2,552-image training pool and a fixed 100-epoch budget, unlike Models A and B, whose parameters are entirely inherited from the pretrained checkpoint. A secondary, non-exclusive possibility is that VNTS signs are already adequately resolved by YOLO26n's default P3 (stride 8) head, in which case the added P2 head introduces competing anchors without addressing a real resolution bottleneck.

VI. DISCUSSION

The central finding of this work is that HLS's apparent benefit does not hold up under a properly separated evaluation protocol: an earlier interim result and an earlier draft of the final results both reported a statistically flavored positive effect for HLS that did not survive a later revision to the evaluation pipeline. Under the corrected protocol — HLS hyperparameters selected via cross-validation on the training pool only, test set evaluated exactly once — the best HLS configuration (HLS-3_hicls) is statistically indistinguishable from a flat-label baseline at every frequency tier, and on raw numbers actually trails the baseline on both global metrics and the Rare tier, only edging ahead on Frequent and Medium by less than one percentage point.

This does not mean HLS hyperparameters are unimportant; quite the opposite. Within the HLS family, classification-loss weight (λ_{cls}) drives a large, low-variance effect across CV folds, roughly doubling Rare-tier performance when moved from 0.5 to 1.5. The honest summary is therefore not "HLS does nothing," but that HLS's own hyperparameters matter a great deal, and even once they are optimized, the result does not clear the bar a plain baseline already sets.

The category-size mechanism hypothesized to explain HLS's effect does not survive contact with the taxonomy's sample sizes: several categories, including Supplementary, contain too few classes for their mean AP50 delta to be distinguished from sampling noise, so the corrected analysis can neither confirm nor rule out a genuine category-size effect — it can only say that the earlier pattern is not reliable evidence for one.

Separately, the P2 architectural enhancement produces a clear, statistically significant negative result — the most confident finding in this work. Its signature is specific rather than diffuse: the highest precision of the three models, paired with a recall collapse and a corresponding drop in the threshold-independent mAP@0.5, points to poorly calibrated confidence from the new, untrained P2 branch, most plausibly because those parameters must learn calibration from scratch within a small-dataset, fixed-epoch budget, rather than a fundamental mismatch between P2 heads and traffic sign detection.

VII. Limitations

- HLS replaces YOLO26n's Task-Aligned Assigner box-quality scaling with a fixed hierarchical target distribution at positive anchors, so Model A's correct-class target no longer reflects localization quality the way Model B's does — a secondary difference beyond the intended hierarchical redistribution, and a candidate confound for future refinement.
- The cross-validation ablation used a 45-epoch budget, constrained by Kaggle's 12-hour per-session runtime limit and 30-hour weekly GPU quota across the ablation's 12 fold-configuration runs, while final training used 100 epochs; the relative ranking among the four HLS configurations is not independently verified at the longer schedule beyond the single winning configuration that was carried forward.
- The internal validation carve-out used for final-model checkpoint selection is small ($\sim$10% of the training pool, roughly 255 images), which may introduce checkpoint-selection noise independent of the underlying model quality.
- A single random seed was used throughout final training; variance is only characterized across CV folds during the ablation phase, not across independent training seeds for the final models.
- The Supplementary category (3 classes, 190 instances) and the Rare tier (13 classes, 260 instances) produce high-variance per-class AP50 estimates regardless of model configuration, limiting confidence in category- and tier-specific conclusions drawn from them.
- Model P2's negative result was not decomposed into its two candidate causes (undertrained new parameters vs. genuine architecture–task mismatch).
- The VNTS dataset contains only 3,216 images; the interaction between either intervention and dataset scale remains untested.

VIII. Future Work

- Repeat Model P2 training with a longer schedule or a staged warm-up (e.g., freezing pretrained layers while the P2 head trains first) to test whether its negative result is a convergence artifact rather than a genuine mismatch with sign object sizes.
- Empirically compare the VNTS bounding-box size distribution against YOLO26n's P2/P3 resolution boundary directly, rather than inferring smallness only from percentage-of-image-area statistics.
- Repeat the HLS cross-validation ablation with multiple random seeds per configuration to separate seed variance from fold variance, and verify the winning configuration's ranking at the full 100-epoch budget.
- Since neither raw instance count ($r = 0.07$) nor QCVN 41 category size predicts HLS's per-class effect, investigate alternative structuring principles (e.g., visual or shape similarity) for a hierarchical target distribution.
- Evaluate on a larger Vietnamese traffic sign dataset to increase statistical power in the Rare tier and Supplementary category.
- Investigate Hierarchical Cosine Loss [8] as an alternative hierarchical formulation; HLS's null result here does not rule out other hierarchical loss designs.

IX. Conclusion

This paper constructed a QCVN 41-aligned hierarchical taxonomy for Vietnamese traffic sign detection and conducted a controlled empirical investigation of two independent interventions against a flat-label YOLO26n baseline: a loss-level hierarchical label smoothing scheme (Model A) and an architecture-level small-object detection head (Model P2). HLS's hyperparameters were selected via cross-validation on the training pool alone, and the test set was evaluated exactly once per final model, a protocol adopted after an earlier pipeline version was found to use the test set as validation during training.

Under this corrected protocol, Model A's CV-selected configuration ($\alpha = 0.05$, $\beta = 0.10$, $\lambda_{cls} = 1.5$) is not statistically distinguishable from the flat baseline at any frequency tier and trails it on raw Global and Rare-tier numbers, even though λ_{cls} has a large, reproducible effect within the HLS family itself: HLS's hyperparameters matter, but tuning them does not close the gap to a flat baseline. The category-size mechanism that appeared to explain HLS's effect under an earlier evaluation protocol can be neither confirmed nor ruled out under the revised per-category analysis, since several categories contain too few classes to separate a genuine effect from sampling noise. Model P2 produces a clear, statistically significant negative result at every tier; its precision/recall/mAP50 signature specifically implicates poorly calibrated confidence in the new, undertrained P2 branch rather than a wholesale loss of detection or classification ability.

Taken together, this work's central contribution is the finding that HLS's apparent benefit does not survive a properly separated evaluation protocol: a plausible-sounding hierarchical loss modification does not outperform a well-tuned flat baseline once hyperparameter selection and final test evaluation are kept independent.

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